CONUS Regional HT-DBH Curves Using FIA Data

Working Draft 3

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1.0 Introduction

This work is part of a project to develop a continental US (CONUS) growth model using Forest Inventory & Analysis (FIA) data for all FIA species without regional variants. The component covered here is the regional tree height (HT-DBH) models. These models are used for missing data and in some cases height growth for minor species.

Regional height growth models are problematic because if used as is they provide bias height eastimates. The reason for this is that regional HT-DBH curves (fit from a collection of plots or stand) tend to have steeper slopes than curves for individual stands (or plots) which tend to have flatten out over time. This bias can be minimized by localizing the regional curves buy calibrating it with tree HT and DBH measurements from the stand (or plot) of interest.

The objective here is to develop regional HT-DBH curves for as many species as possible using a consistent equation for all species that is easily calibrated by users.

2.0 Methods

The following models were selected for fitting as they are widely used and have good performance in published comparisons (e.g., Curtis 1967; Huang and Titus (1992), Zhang 1997, Martin and Flewelling 1998).

$$HT = bh + exp(B1 + B2 * DBH^{-1.0}) \tag{1}$$

$$HT = bh + exp(B1 + B2 * DBH^{B3}) \tag{2}$$

$$HT = bh + B1 * exp(1.0 - B2 * DBH)^{B3} \tag{3}$$

$$HT = bh + exp(B1 + B2/(DBH + 1.0)) \tag{4}$$

$$HT = bh + exp(B1 + B2/(DBH + B3)) \tag{5}$$

$$HT = bh + B1 * (1.0 - exp(B2 * DBH))^{B3} \tag{6}$$

where bh is the breast height (4.5 feet), HT is total height (feet) and DBH is diameter at breast height (inches).

Models 1 and 2 are a Schumacher (1939) type curves (Curtis 1967). Model 3 is the Chapman-Richards function. Models 4 and 5 are often used in FVS (Wykoff and others 1982). Model 6 is a Weibull type curve.

Each model was fit by species that had at least 3 observations. Fits were weighted by 1/DBH (based on residual plots and published results) and were fit using the "nls" function in R with 150 interation limit. The fits were evaluated for reasonable parameters (correct signs), low root mean square error (RMSE) and minimum bias.

The data is all live FIA trees with a measured HT (filtered bu HTCD=1) and DBH greater than 0 and HT greater then 4.5. This provided 9,509,054 observations for 438 species (excluding AK).

3.0 Results

The table below summarizes the number of species fit and no fit for models 1 through 6.

- Nspecies = total number of species
- Nfit = number of species fit
- Nminobs = number of species that did not have the required minimum number of trees
- Nnofit = number of species that did not successfully fit (convergence, maximum interates, etc.)
- Npoorcoef = number of fits that had coefficients that were the wrong sign
- N__Bns = number of fits with nonsignificant (Ho=0) parameters
- N__Bgood = number of "good" fits
- N__B3test = number of fits that the B3 parameter was not different from -1 (model 2) or 1 (model 4).

Significance was tested at the alpha 0.05 level which is is arbitrary since there might be justification of accepting fits with higher alpha levels. Also, for models 2 and 5 it may not actually be important of the B3 parameter is different form -1 or 1, respectively, but is given here mostly for interest.

Table 1: Summary of fits by model (1-6).

model	1	2	3	4	5	6
Nspecies	438	438	438	438	438	438
Nfit	385	338	239	386	361	254
Nminobs	26	26	26	26	26	26
Nnofit	NA	74	173	NA	34	158
Npoorcoef	NA	NA	NA	1	17	NA
N__Bns	27	48	24	25	64	22
N__Bgood	407	308	237	406	354	250
N__B3test	NA	296	NA	NA	260	NA

The table below summarizes the resulting mean RMSE and mean bias for models 1 through 6. Note that there are different numbers of fits for each model (see table above) so comparisons need to be done with some caution.

- minRMSEcount = the number of species fits that had the lowest RMSE
- meanRMSE = the unweighted mean RMSE for successful fits (all species)
- meanRMSEw = the weighted mean RMSE for successful fits (all species)
- minBIAScount = the number of species fits that had the lowest bias
- meanBIAS = the unweighted mean bias for successful fits (all species)
- meanBIASw = the weighted mean bias for successful fits (all species)

Table 2: Summary of fit statistics by model (1-6).

model	1	2	3	4	5	6
minRMSEcount	16	84	60	42	140	50
meanRMSE	3.45	3.26	3.34	3.26	3.22	3.29
meanRMSEw	3.89	3.51	3.49	3.65	3.5	3.55

minBIAScount	8	111	77	19	73	104
meanBIAS	-0.7058	0.0129	-0.0011	-0.317	-0.0101	0.0011
meanBIASw	-0.9126	0.0121	0.0001	-0.4345	-0.0157	-0.0007

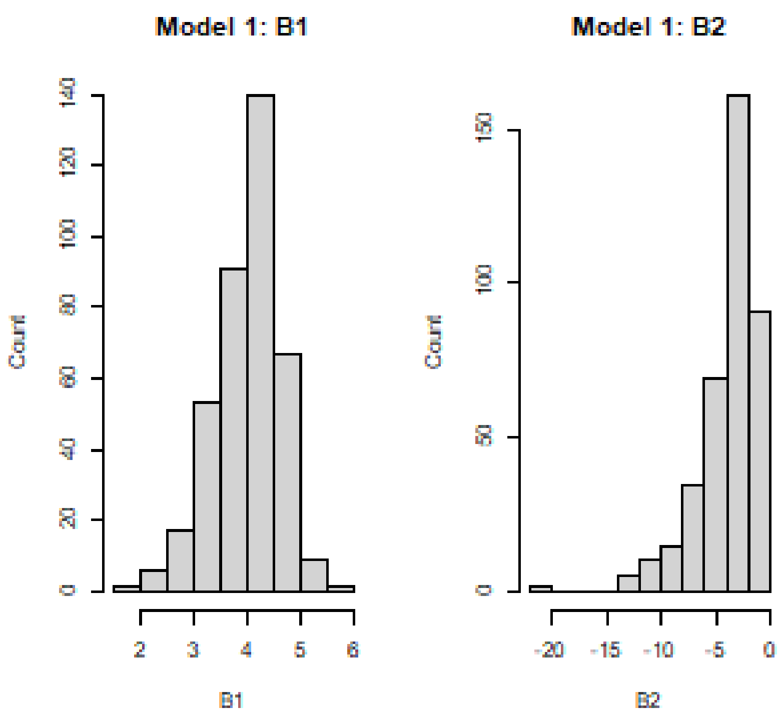
The table below summarizes the mean (unweighted) bias for the lower, middle and upper third of the DBH distributio of each species fit. This may give some idea of how the models fit the small and larger trees.

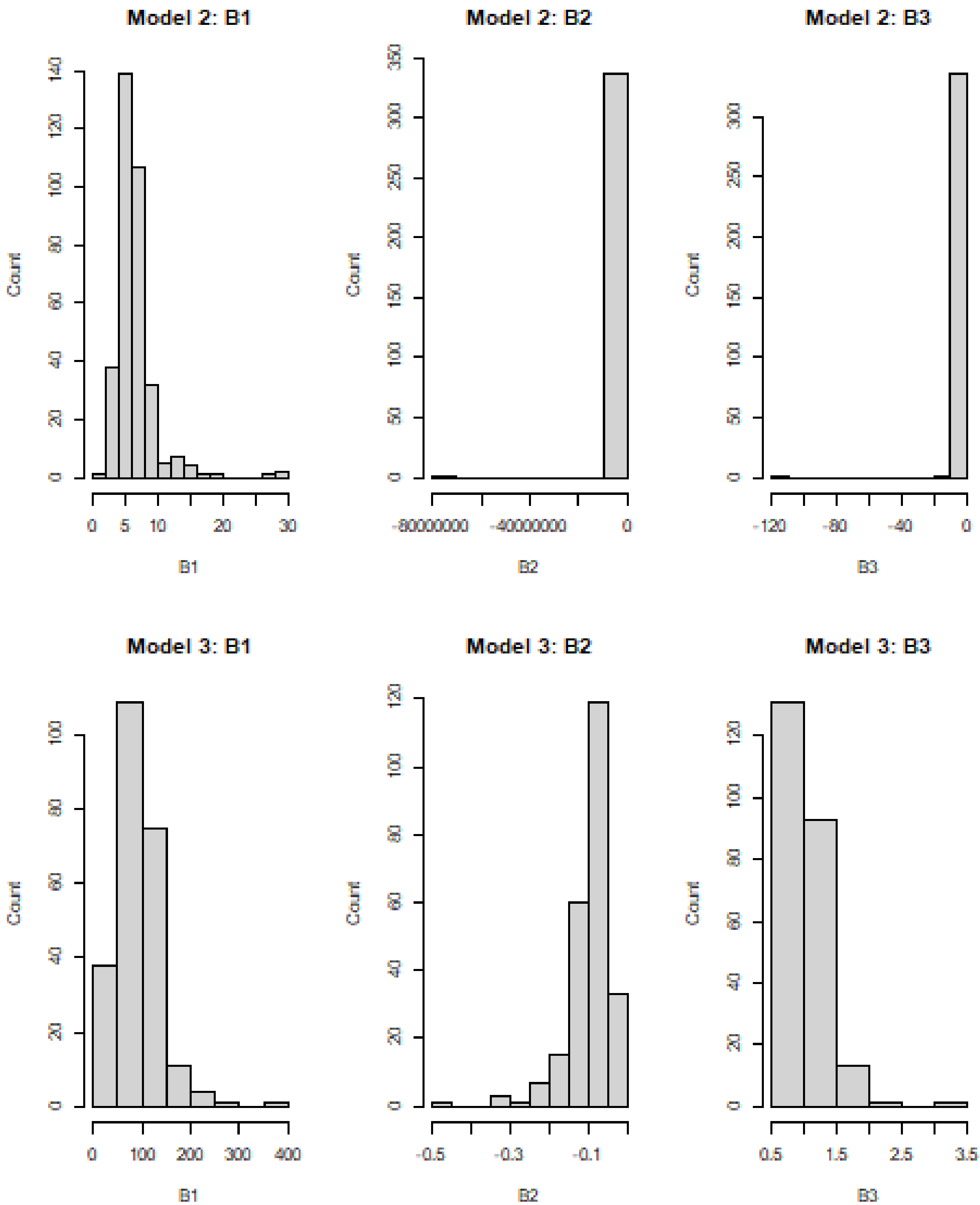
minLOWcount = the number of species fit with the lowest mean bias in the lower third of the diameter distribution by model meanBIASlow = the mean bias for the lower third of the diameter distributions minMIDcount = the number of species fit with the lowest mean bias in the middle third of the diameter distribution by model meanBIASmid = the mean bias for the middle third of the diameter distributions minUPPcount = the number of species fit with the lowest mean bias in the upper third of the diameter distribution by model meanBIASUPP = the mean bias for the upper third of the diameter distributions

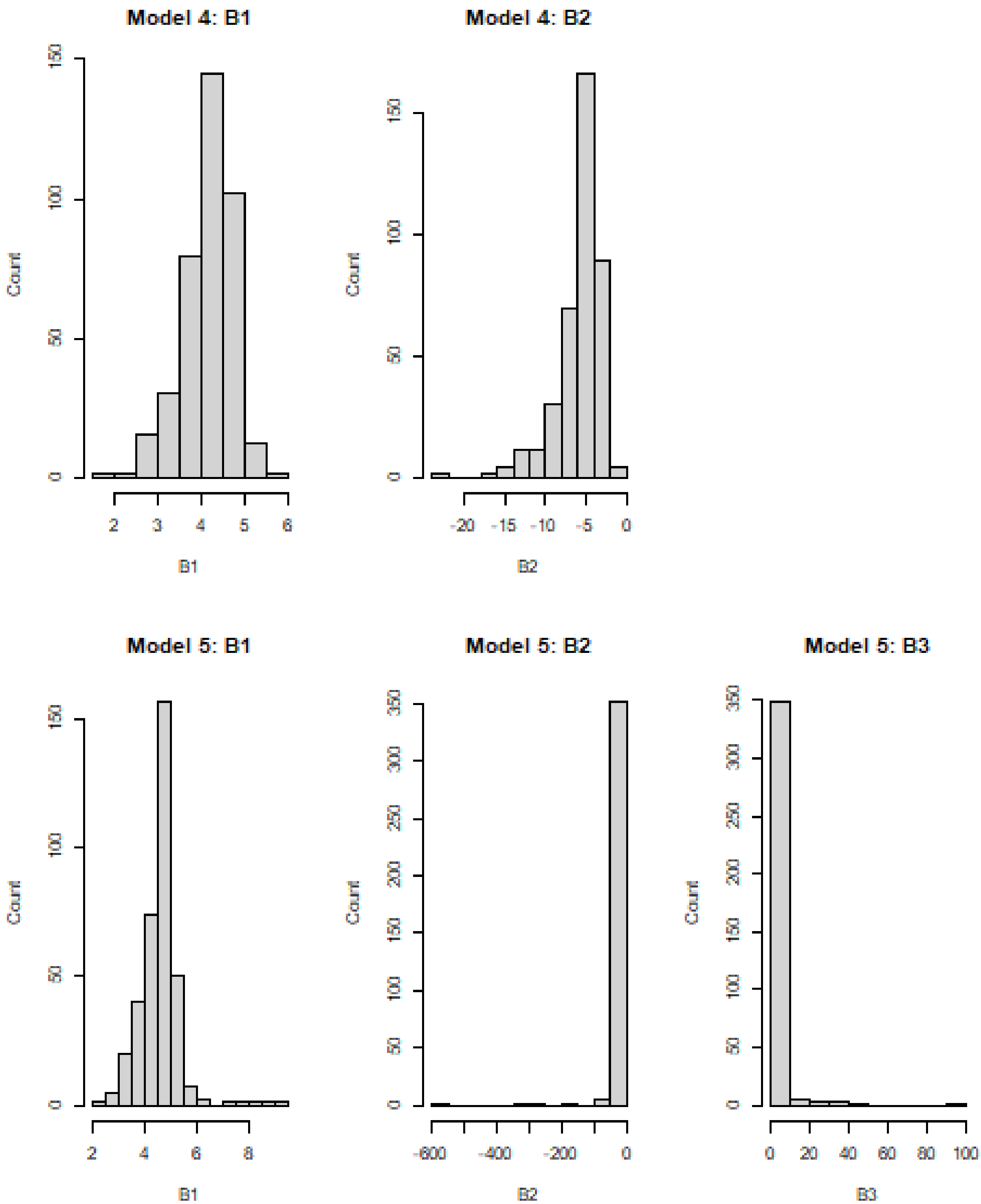
Table 3: Summary of number of lowest bias models and the mean bias in the lower, middle and upper thirds of the DBH distributions by model (1-6).

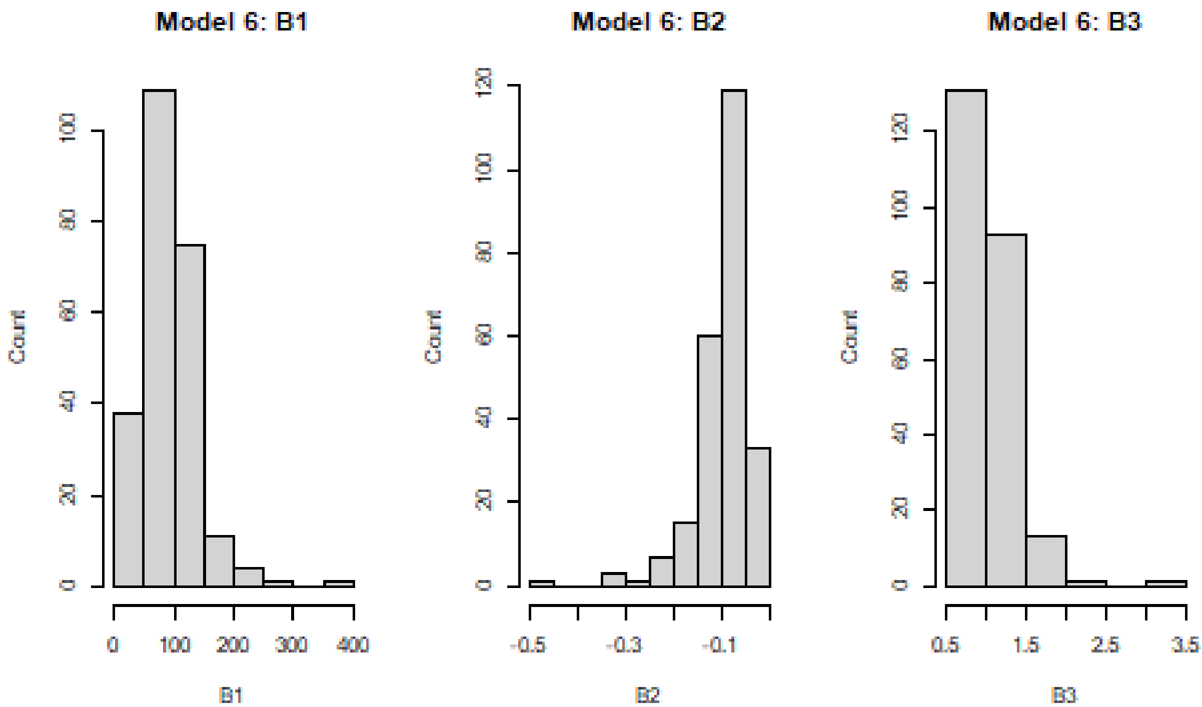
model	1	2	3	4	5	6
minLOWcount	49	78	35	50	134	46
meanBIASlow	0.135	0.177	0.050	0.066	-0.047	0.036
minMIDcount	36	108	20	47	129	52
meanBIASmid	1.421	-0.188	-0.093	0.987	0.073	-0.102
minUPPcount	15	87	46	36	156	52
meanBIASUPP	-3.687	0.050	0.041	-2.010	-0.054	0.070

Below are histogram plots of the coefficients for all “successful” (convergence) fits each model.

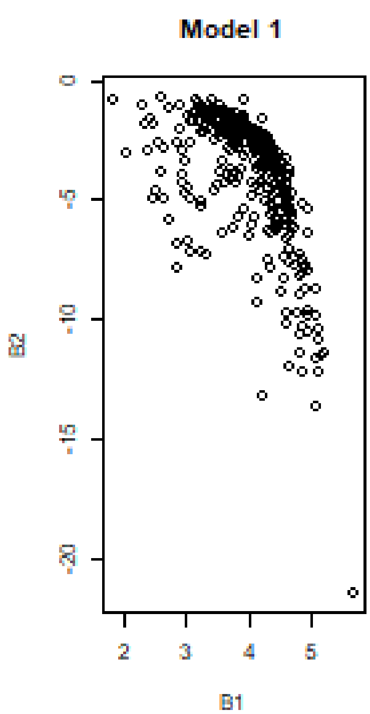


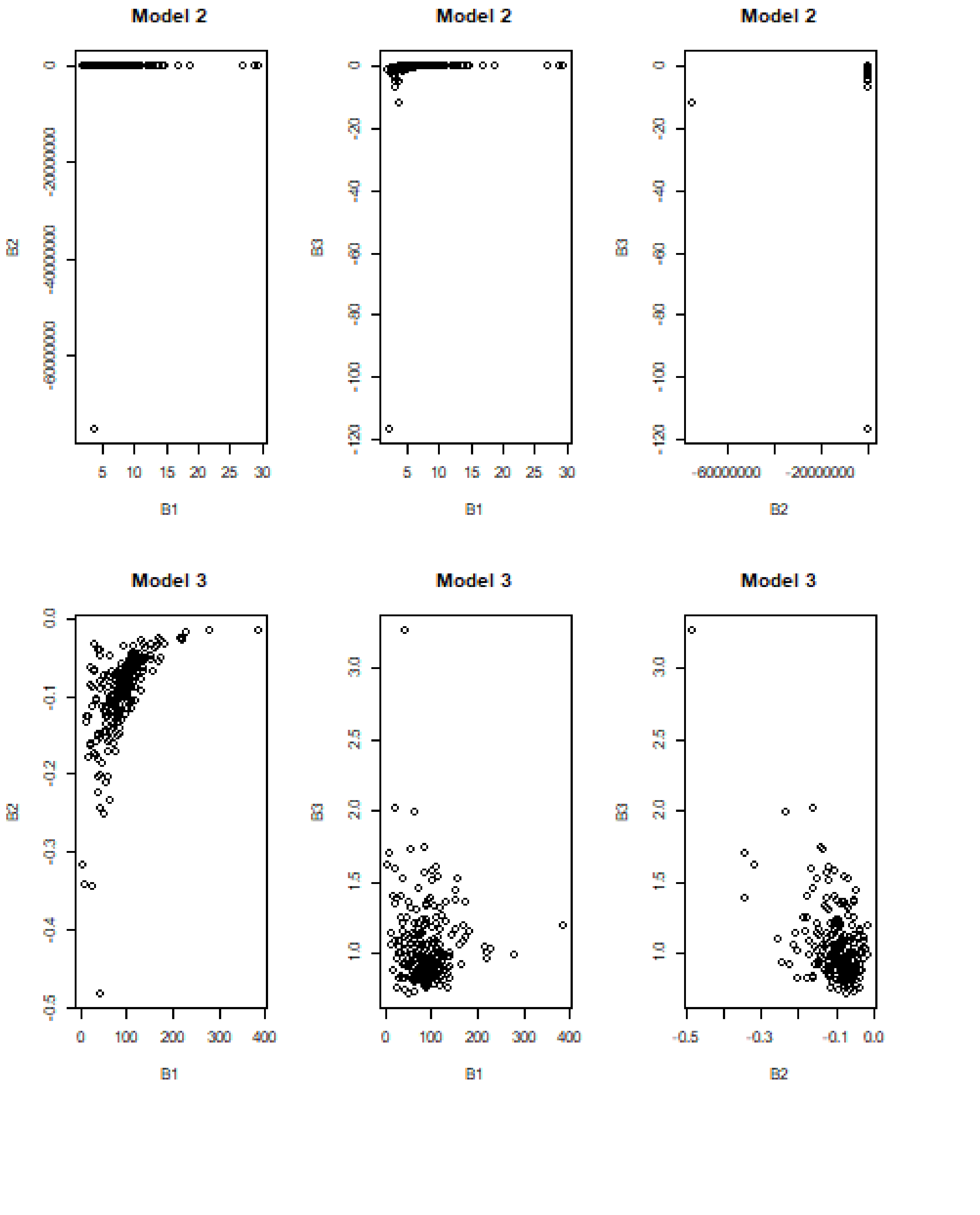

Model 2: B1
Model 2: B2
Model 2: B3
Count
B1
B2
B3
Model 3: B1
Model 3: B2
Model 3: B3
Count
B1
B2
B3

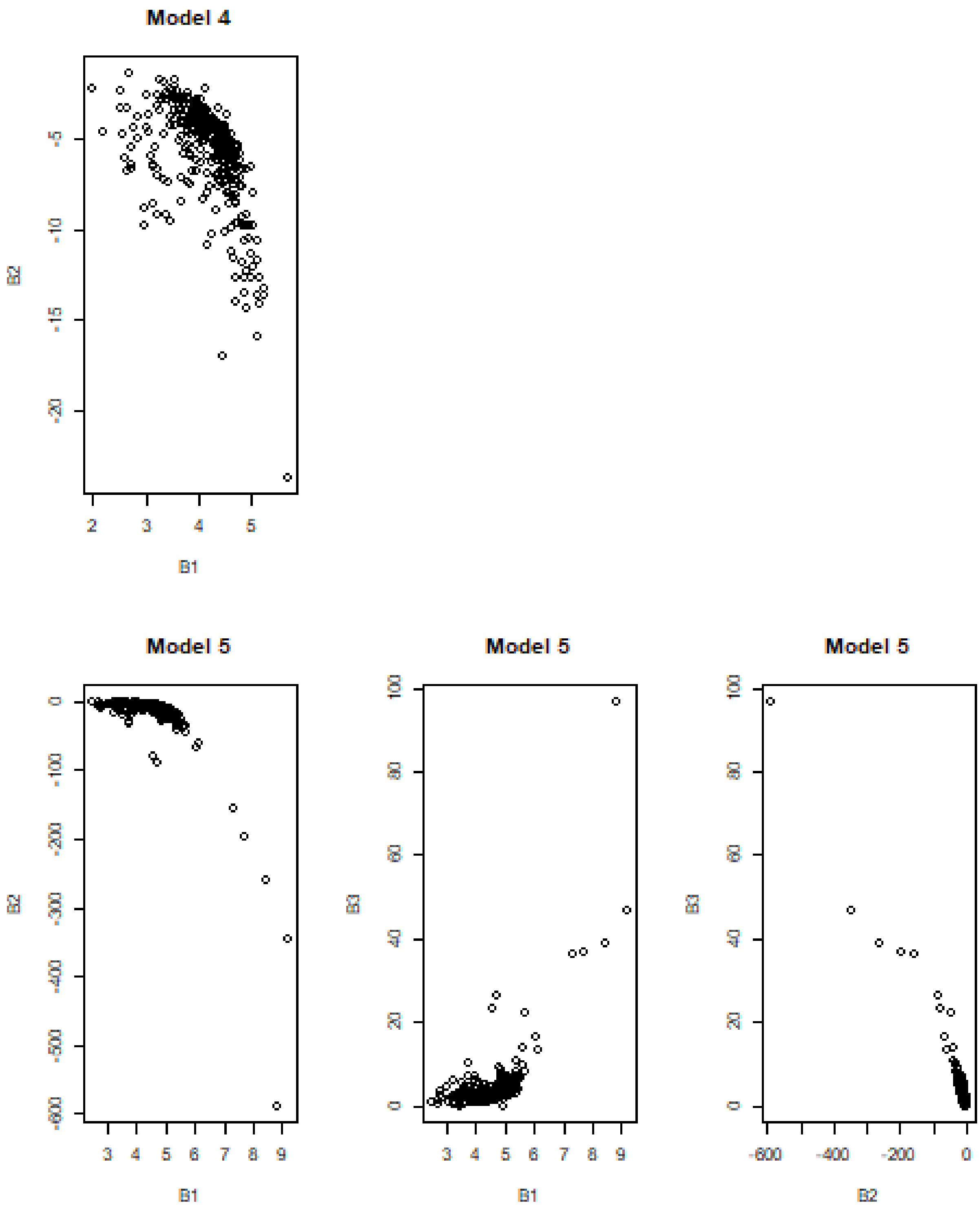

Model 4: B1
Count
B1
0
50
100
150
2
3
4
5
6
Model 4: B2
Count
B2
0
50
100
150
-20
-15
-10
-5
0
Model 5: B1
Count
B1
0
50
100
150
2
4
6
8
Model 5: B2
Count
B2
0
50
100
150
200
250
300
350
-600
-400
-200
0
Model 5: B3
Count
B3
0
50
100
150
200
250
300
350
0
20
40
60
80
100

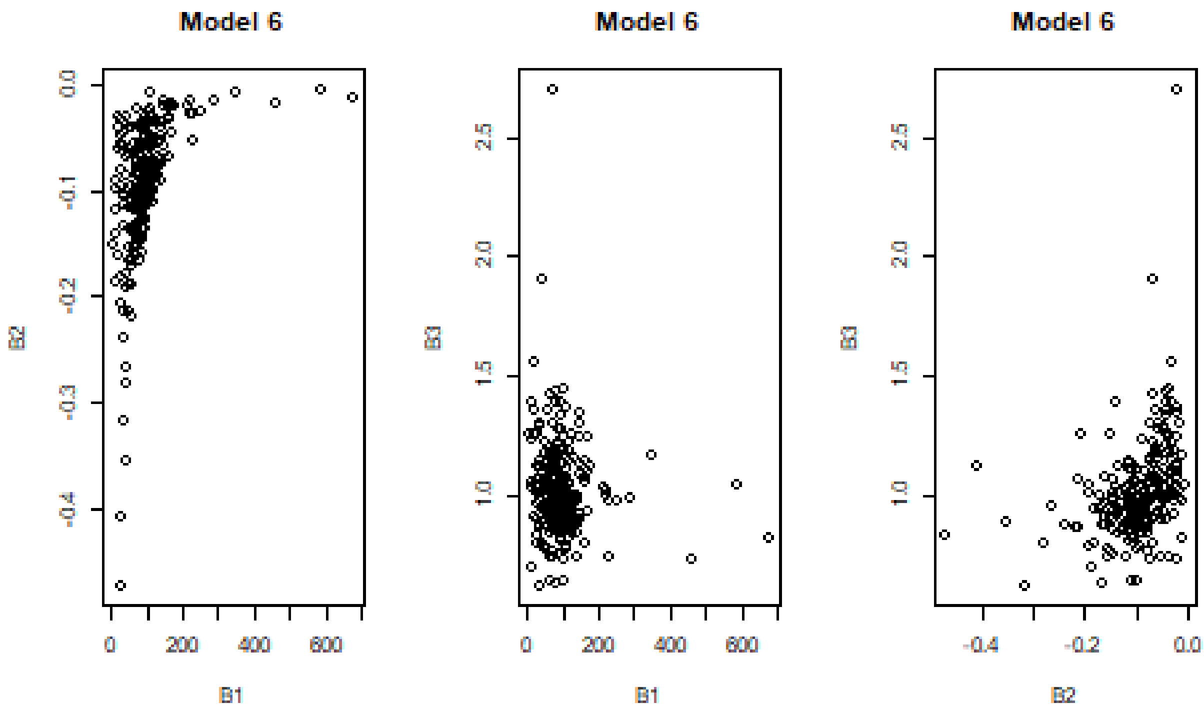


Below are bivariate plots of the coefficients for all "successful" (convergence) fits each model.



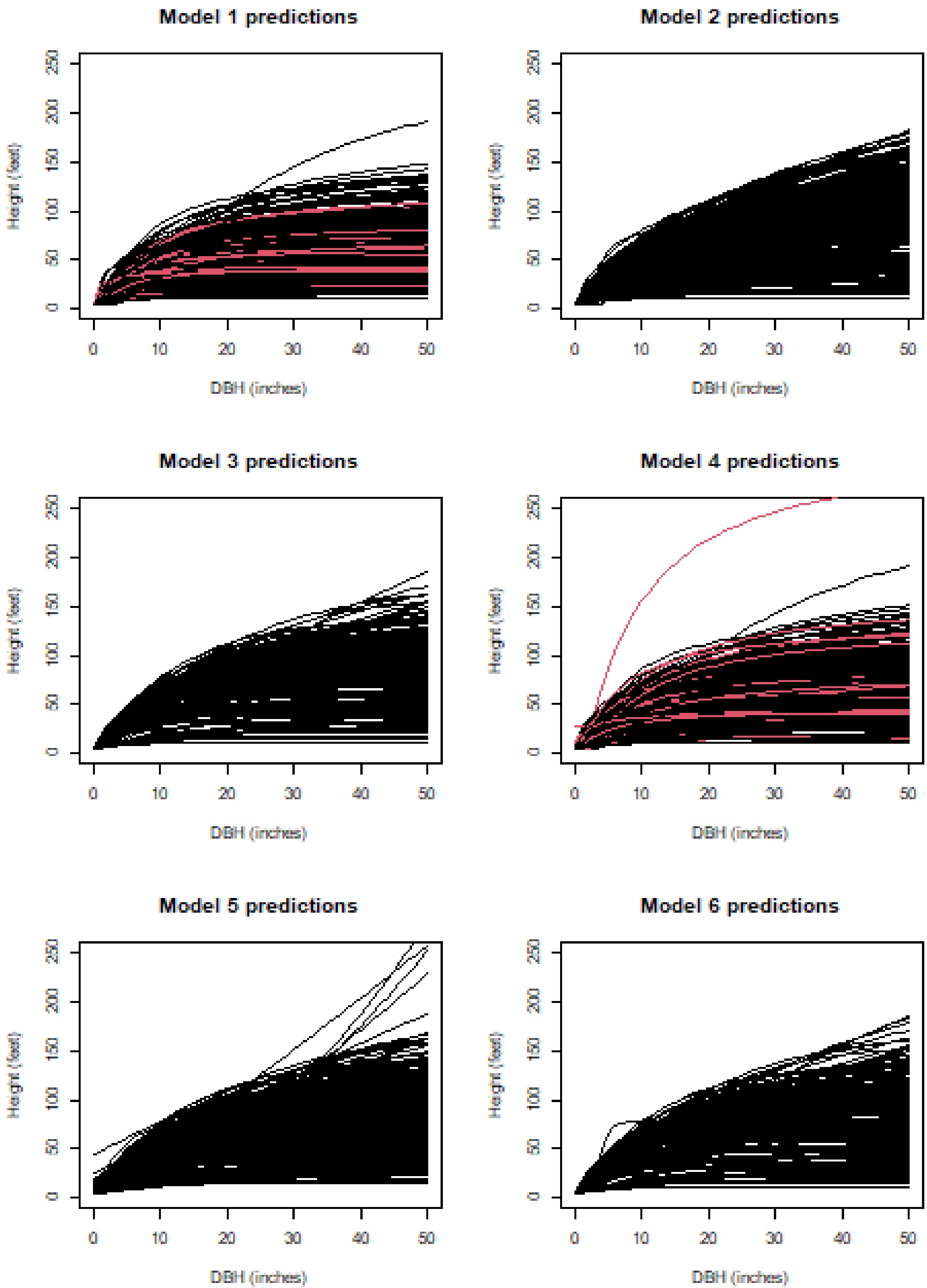

Model 2
B2
B1
Model 2
B3
B1
Model 2
B3
B2
Model 3
B2
B1
Model 3
B3
B1
Model 3
B3
B2


Model 4
B1
B2
Model 5
B1
B2
Model 5
B1
B3
Model 5
B2
B3



Finally, the graphs below are the predictions for all "successful" (convergence) fits each model. The red colored lines indicate fits that converged but at least one of the parameters might be insignificant (not different from 0).

All of the fits that had a predicted height of 160 or 170 feet at a DBH of 50 inches was plotted against the raw data. In all cases the curves appeared to fit the range of the data. In some cases the data set for a species was small and/or nearly linear. In some cases the range of the DBH as limited and did not include larger (e.g. greater than 18-inch) trees and did not exhibit an asymptotic behavior.


Model 1 predictions
Model 2 predictions
Model 3 predictions
Model 4 predictions
Model 5 predictions
Model 6 predictions
Height (feet)
DBH (inches)
0
50
100
150
200
250
0
10
20
30
40
50

4.0 Discussion and Next Steps

The results here are preliminary but are a start at building library of HT-DBH curves for all FIA species.

These fits were done with nonlinear least squares, however, give that plots have remeasurements (within plot correlation structure) one might question of a mixed models (MM) with random effects for plots should have been used. This approach would account for the plot and remeasurement correlation. However, it would not solve the issue of a regionally fit curve being biased for individual stands (plots) without some type of calibration. This can be done but requires measured trees from the stand (plot) if interest and the calculations are not straight forward and requires using the variance/covariance structure to recover estimate of the random effects. The models above can be calibrated using measured trees in a bit more straight (ordinary least square type approach) forward manner as (Temsegen and others 2008):

$$K = sum((pHT - 4.5) * (HT - 4.5))/sum((pHT - 4.5)^2)$$

where K is the height calibration, pHT is the predicted height, and HT is a measured height. The calibration is applied as

$$adjHT = 4.5 + K * (pHT - 4.5)$$

Temesgen and others (2008) compared calibrations using OLS and MM approaches and (my opinion) showed little differences (maybe a few less trees need for the MM approach), but both required tree measurements and the MM approach was less straight forward to implement.

The big questions are (a) which model to use and (b) which fits to not use?

For the first the as we might expect from the results of other model comparisons the differences are not great (models tend to go through the middle of the data). The Weibull model (6) is probably at the bottom. Models 1 and 5 get the most "good" fits and are the most linear like. Many of the fits for models 2 and 5 have B3 estimates that are not different from -1 and 1 (respectively), which may or may not matter.

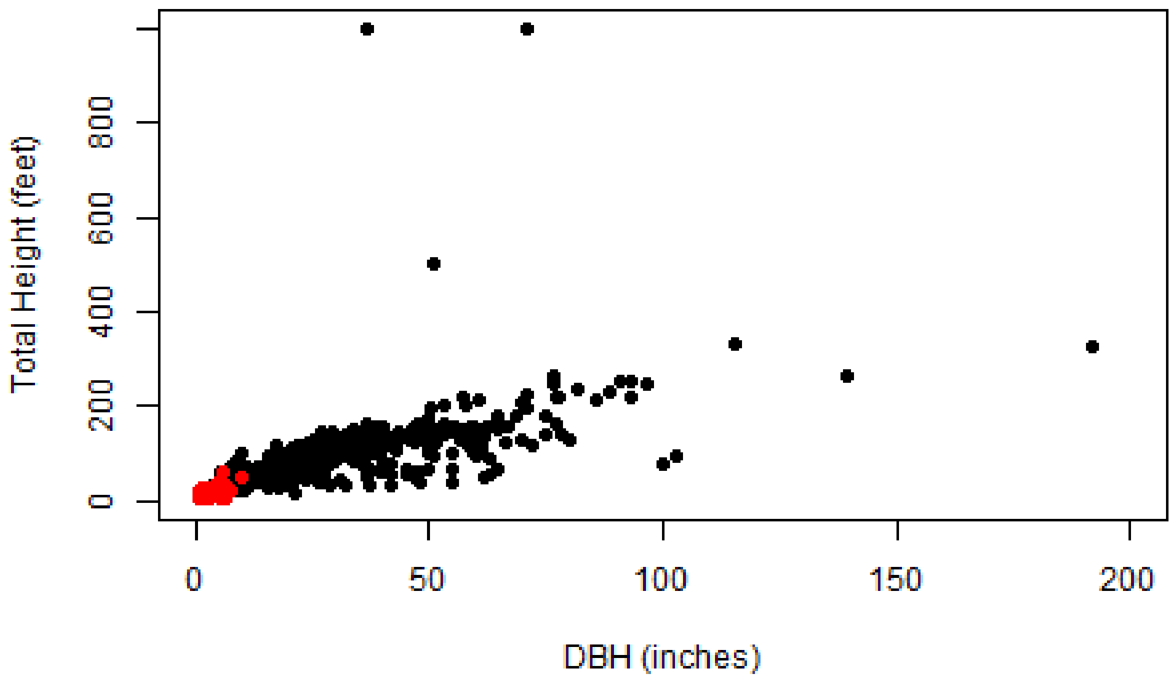
Models 2, 3 and 5 tended top have the most species with the lowest RMSE and the lowest average RMSE and model 5 had the lowest mean bias across the DBH classes. However, model 5 had some species with strange fits (high values for small diameters and non-asymptotic behavior) which I am not sure why. Model 2 was also good but over-predicted the small trees but was very good at the middle and larger trees in the DBH distribution.

Considering which fits not to use, migtht have more to do with the data than the models. The graph below shows the minimum DBH and HT (red) and the maximum DBH and HT (black) for each species. Note that some of these species have very small maximum DBHs. Is thisa characteristic of the species or the sample?

These graphs also show that there is a need for some data filtering for extreme and unreasonable values (see below).

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.10   14.93   26.45   31.19   41.88  192.00
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     9.0    52.0    94.5   100.6   130.0   999.0
```



Some things still to do . . .

1. There appear to be more "outlier" heights and diameter than I was expecting. I don't know how much difference these make, but some could be influential (model 5 shapes?) and probably need to be addressed. With 438 species this could take a while unless some rules could be used (e.g. HT/DBH ratio limits).

2. I have used "nls" in R but should try "nlsLM". This should not change the overall results here but "nlsLM" tends to be a bit more robust and might (maybe) converge on a few more species. This is a fairly quick check to do.

3. Will need to develop species mapping. I started looking at the species group values (SPGRPCD) to see if this could help with this. There are 48 groups (after dropping the islands) for East, West, hardwoods and softwoods.

5.0 References

Curtis, R.O. 1967. Height-diameter and height-diameter-age equations for second-growth Douglas-fir. Forest Science 14(4):365-375.

Huang, S.; Titus, S. 1992. Comparison of nonlinear height-diameter functions for major Alberta tree species. Canadian Journal of Forest Research 22:1297-1304.

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